

Chapter 22 - Proteobacteria

Bacteria Name	Characteristics
<i>Rhodospirillum</i> (genus)	Purple nonsulfur bacteria, produces H ₂ , novel biodegradable plastic, and oxidize carbon monoxide to carbon dioxide
<i>Rickettsia</i> (genus)	All species are parasitic or mutualistic <i>Rickettsia prowazekii</i> and <i>Rickettsia typhi</i> cause typhus fever <i>Rickettsia rickettsii</i> cause Rocky Mountain Spotted Fever
<i>Caulobacteraceae</i> (family)	Budding bacteria Reproduction: asymmetric transverse binary fission
<i>Hyphomicrobiaceae</i> (family)	Prosthecae and budding bacteria
<i>Rhizobiaceae</i> (family) <i>Rhizobium</i> (genus)	Pleomorphic Grow symbiotically as nitrogen-fixing bacteroids within root nodule cells of legumes
<i>Rhizobiaceae</i> (family) <i>Agrobacterium</i> (genus)	Invade crown, roots, and stems of many plants transform infected plant cells into proliferating tumors <i>Agrobacterium tumefaciens</i> causes crown gall disease by means of tumor-inducing (Ti) plasmid
<i>Brucella</i> (genus)	Causes undulant fever
methylophilic <i>Methylocystis</i> and <i>Methylosinus</i> (genra)	Use 1-carbon compounds for carbon and energy first step is oxidation to formaldehyde second step is either incorporation (biomass) or further oxidation (energy) Possess extensive intracellular membranes Often form a resting cell called exospore
<i>Nitrospira</i> (genus) & <i>Nitrosococcus</i> (genus)	Oxidize ammonia to nitrite (nitrification)
<i>Nitrobacter</i> (genus) & <i>Nitrococcus</i> (genus)	Oxidize nitrite to nitrate (nitrification)
<i>Magnetococcus</i> (genus) <i>M. marinus</i> (species)	Magnetotaxis mediated by single row of magnetite crystals arranged in cytoplasm
<i>Pelagibacter ubique</i>	sole cultured representative of the group SAR11 Clade, the most abundant group of microbes on earth
<i>Neisseria</i> (genus)	Inhabitants of mucous membranes of mammals some human pathogens: <i>Neisseria gonorrhoeae</i> – gonorrhea <i>Neisseria meningitidis</i> – meningitis
<i>Burkholderiales</i> (order)	Some members have a sheath, hollow tubelike structure surrounding chain of cells.

<i>Burkholderia</i> (genus)	Most use poly-b-hydroxybutyrate as carbon reserve Form symbiotic associations with legumes similar to that formed by rhizobia
<i>Burkholderia cepacia</i>	Plant pathogen Nosocomial pathogen particular problem for cystic fibrosis patients
<i>Ralstonia</i> (genus)	Form symbiotic associations with legumes similar to that formed by rhizobia
<i>Bordetella</i> (genus)	Mammalian parasites that multiply in respiratory epithelial cells. They cause whooping cough and kennel cough
<i>Nitrosomonadales</i> (order) <i>Nitrosomonas</i> and <i>Nitrospira</i> (genra) <i>Nitrosomonas europaea</i>	Oxidize ammonia to nitrite
<i>Thiobacillus</i> (genus)	Oxidize elemental sulfur, hydrogen sulfide, and thiosulfate to sulfate while generating energy
<i>Chromatiales</i> (order) <i>Chromatiaceae</i> & <i>Ectothiorhodospiraceae</i> (families)	Purple sulfur bacteria that oxidize hydrogen sulfide to sulfur and deposit the granules internally
<i>Leucothrix</i> (genus)	Forms gonidia
<i>Thiothrix</i> (genus)	Related to <i>Leucothrix</i> with gonidia
<i>Beggiatoa</i> , <i>Thioploca</i> , and <i>Thiomargarita</i>	Among the largest microbes. They grow in bundles
<i>Methylococcaceae</i> (family)	Almost all form resting stage (cystlike structure)
<i>Legionella pneumophila</i>	Causative agent Legionnaire's disease Intracellular pathogen of protozoa Invade cooling towers, air conditioning, hot tubs
<i>Pseudomonas</i> (genus)	Participate in natural mineralization processes Spoil refrigerated food
<i>Shewanella</i> (genus)	Form nanowires
<i>Vibrio cholerae</i>	Pathogen that causes cholera Has two circular chromosomes, cholera toxin gene is integrated phage on chromosome 1
<i>V. fischeri</i> , & <i>V. harveyi</i>	Capable of bioluminescence Emission of light catalyzed by luciferase Symbiotic relationship with fish organs, and some <i>Photobacterium</i>
<i>Escherichia coli</i>	Inhabitant of intestinal tracts of many animals Used as indicator organisms for testing water for fecal contamination Some strains are pathogenic and cause: - Gastroenteritis

	- Urinary tract infections
<i>Salmonella</i> (enteric bacteria)	Causes typhoid fever and gastroenteritis
<i>Shigella</i> (enteric bacteria)	Causes bacillary dysentery
<i>Klebsiella</i> (enteric bacteria)	Causes pneumonia
<i>Yersinia</i> (enteric bacteria)	Causes plague
<i>Erwinia</i> (enteric bacteria)	Causes blights, wilts, etc., of crop plants
<i>Pasteurella multocida</i>	Causes fowl cholera
<i>Pasteurella haemolytica</i>	Causes pneumonia in cattle, sheep, and goats
<i>Haemophilus influenzae</i>	Causes meningitis in children
<i>Desulfuromonales</i> (order)	Conserve energy from dissimilatory metal reduction including toxic and radioactive metals Electricity generated by microbial fuel cells Synthesize nanowires
<i>Bdellovibrio</i> (genus)	Predatory bacteria life cycle that resembles bacteriophages in many ways
<i>Myxococcales</i> (order)	Have complex life cycle that leads to the production of dormant myxospores held within fruiting bodies
<i>Myxococcales xanthus</i>	Have - Social (S) motility – twitching motility - Adventurous (A) motility
<i>Campylobacter fetus</i>	Causes reproductive disease and abortions in cattle and sheep Causes septicemia, enteritis and Guillain-Barre syndrome in humans
<i>Campylobacter jejuni</i>	Causes abortions in sheep, and enteritis diarrhea in humans
<i>Helicobacter pylori</i>	Causes gastritis and peptic ulcer diseases Does not grow below pH 4.5 Urease converts urea to ammonia and CO ₂

Chapter 23 - Firmicutes

Bacteria Name	Characteristics
<i>Clostridium</i> (genus)	Ferment amino acids using Stickland reaction (l sodium motive force)
<i>Clostridium botulinum</i>	Causes food spoilage, and botulism disease
<i>Clostridium tetani</i>	Causes tetanus disease
<i>Clostridium perfringens</i>	Causes gas gangrene
<i>Clostridium acetobutylicum</i>	Manufacture of butanol
<i>Desulfotomaculum</i>	Stains Gram-negative but in electron micrographs is seen to have a Gram-positive cell wall
<i>Heliobacterium</i> & <i>Heliophilum</i> (genre)	Have bacteriochlorophyll g
Negativicutes (class)	Gram-Positive Bacteria with Outer Membranes
<i>Negativicoccus succinicivorans</i>	Was isolated from human skin and soft-tissue wounds in 2010
<i>Veillonella</i> (genus)	Normal biota of mouth, GI tract, urogenital tract of humans and animals Found in infections of head, lung, female genital tract
<i>Bacillus subtilis</i>	Used as model organism for gene regulation, cell division, quorum sensing, cellular differentiation Form biofilms and fruiting bodies
<i>Bacillus cereus</i>	Causes food poisoning
<i>Bacillus anthracis</i>	Causes anthrax
<i>Bacillus thuringiensis</i> and <i>Bacillus sphaericus</i>	Used as insecticide
<i>Thermoactinomyces</i>	Found in high temperature environments such as composts Causes farmer's lung
<i>Sporosarcina</i> (genus)	Only known endospore-former that has cocci and not rod shape Tolerates pH up to 10. Degrades urea to ammonia and carbon dioxide
<i>Staphylococcus epidermidis</i>	Common skin resident Causes endocarditis
<i>Staphylococcus aureus</i>	Most important human staphylococcal pathogen; e.g. abscesses, boils, wound infections, pneumonia, toxic shock syndrome, and food poisoning

	Virulence factors: coagulase which causes blood plasma to clot the toxin β -hemolysin lyses cells
Methicillin-resistant <i>S. aureus</i> (MRSA) and vancomycin resistant <i>S. aureus</i> (VRSA)	Antibiotic resistant
<i>Listeria monocytogenes</i>	Causes listeriosis, food poisoning
<i>Lactobacillus</i> (genus)	Carry out lactic acid fermentation
<i>Leuconostoc</i> (genus)	Carry out heterolactic fermentation using the phosphoketolase pathway Involved in production of fermented vegetables and dairy products
<i>Streptococcus pyogenes</i>	Causes streptococcal sore throat, acute glomerulonephritis, and rheumatic fever
<i>Streptococcus mutans</i>	Causes dental caries
<i>Streptococcus pneumoniae</i>	Causes lobar pneumonia and otitis media
<i>Enterococcus faecalis</i>	Normal biotic in gastrointestinal tract Opportunistic pathogen (urinary tract infections and endocarditis)
<i>Lactococcus lactis</i>	Production of buttermilk and cheese

Chapter 24 - Actinobacteria

Bacteria Name	Characteristics
<i>Actinomycetes</i>	Form branching hyphae and asexual spores Source of most currently used antibiotics
<i>Actinomyces</i> (genus)	Cause lumpy jaw in cattle, and ocular infection, actinomycoses, and periodontal disease in humans
<i>Arthrobacter</i> (Genus)	Rod-coccus growth cycle: - exponential phase irregular, branched rods reproduce by snapping division - early stationary phase begin to change to coccoid form
<i>Corynebacterium diphtheriae</i>	Causes diphtheria
<i>Mycobacteriaceae</i> (family)	Contain waxes (mycolic acids) Mycolic acids linked to peptidoglycan by arabinogalactan, a polysaccharide Cell walls very hydrophobic; impenetrable by antibiotics
<i>Mycobacterium bovis</i>	Causes tuberculosis in cattle, humans
<i>Mycobacterium tuberculosis</i>	Causes tuberculosis in humans
<i>Mycobacterium leprae</i>	leprosy
<i>Nocardia</i> (genus)	Develop a substrate mycelium that readily breaks into rods and coccoid elements Some also form an aerial mycelium and conidia Some are opportunistic pathogens causing nocardiosis
<i>Micromonosporales</i> (order)	Includes bacteria that produce morphologically diverse spores
<i>Propionibacterium acne</i>	Skin inhabitant, develops body odor and acne vulgaris
<i>Streptomyces</i>	Produce many antibiotics
<i>Streptomyetales</i> (order)	Aerial hyphae that divide in single plane to form chains of spores
<i>Streptomyces</i> (genus)	produce geosmin; a volatile substance that is source of moist earth odor
<i>Streptomyces coelicolor</i>	One of the largest bacterial genomes, 8.67 Mbp
<i>Streptomyces scabies</i>	Causes scab disease in potatoes and beets
<i>Streptomyces somaliensis</i>	Causes actinomycetoma; infection of subcutaneous tissues in humans
<i>Frankiales</i> (Order)	Fix N ₂ in trees (nonleguminous plants)
<i>Gardnerella</i> (genus)	Causes vaginitis

What is required from chapter 35 tables

Table 35.1 Examples of Pathogenicity Islands and the Products They Encode

Organism	Pathogenicity Island	Gene Product	Function
<i>Helicobacter pylori</i>	PAI-III	Type 1 pili Secreted protein	Attachment Hemolysin, cytotoxic necrotizing factor, uropathogenic protein
<i>Salmonella enterica</i>	SPI-1, SPI-2	Type III secretion Protein	Cytotoxin
<i>Staphylococcus aureus</i>	SaPI	Secreted proteins	Superantigens

Table 35.2 Examples of Microbial Attachment mechanisms

Microbe	Disease	Adhesion Mechanism	Host Receptor
<i>Escherichia coli</i>	Diarrhea	Type I fimbriae	Sugar residue on intestinal epithelium
<i>Streptococcus pyogenes</i>	Sore throat	Protein F	Protein residue on upper respiratory tract cell
<i>Influenza virus</i>	Influenza	Hemagglutinin spike protein	Protein residue on upper respiratory tract cell

Table 35.4 Exotoxins Produced by Human bacterial Pathogens

Toxin	Organism	Gene Location	Toxin Type	Mechanism of Action
Botulinum toxin	<i>Clostridium botulinum</i>	Prophage	AB	Blocks neurotransmitter release, leading to paralysis
Streptolysin O Erythrogenic toxin	<i>Streptococcus pyogenes</i>	Chromosome Phage	Cytolysin superantigen	Hemolysis Cytokine- induced shock
Cholera toxin	<i>Vibrio cholerae</i>	Phage	AB	↑cAMP, leading to water secretion from cell

All the information in table 35.3 are required.